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Determinants of Humanities and Social Sciences Students' Intentions to Use Artificial Intelligence Applications for Academic Purposes

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Abstract: Recent research emphasizes the importance of Artificial Intelligence applications as supporting tools for students in higher education. Simultaneously, an intensive exchange of views has started in the public debate in the international educational community. However, for a more proper use of these applications, it is necessary to investigate the factors that explain their intention and actual use in the future. With the Unified Theory of Acceptance and Use of Technology (UTAUT2) model, this work analyses the factors influencing students' use and intention to use Artificial Intelligence technology. For this purpose, a sample of 197 Greek students at the School of Humanities and Social Sciences from the University of Patras participated in a survey. The findings highlight that expected performance, habit, and enjoyment of these Artificial Intelligence applications are key determinants influencing teachers' intentions to use them. Moreover, behavioural intention, habit, and facilitating conditions explain the usage of these Artificial Intelligence applications. This study did not reveal any moderating effects. The limitations, practical implications, and proposed directions for future research based on these results are discussed.

Keywords: artificial intelligence; behavioural intentions; actual use; higher education; UTAUT2 technology acceptance model

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Citation: Lavidas, K.; Voulgari, I.; Papadakis, S.; Athanassopoulos, S.; Anastasiou, A.; Filippidi, A.; Komis, V.; Karacapilidis, N. Determinants of Humanities and Social Sciences Students' Intentions to Use Artificial Intelligence Applications for Academic Purposes. *Information* **2024**, *15*, 314. https://doi.org/10.3390/info15060314

Academic Editors: Petros Lameras, Sylvester Arnab and Panagiotis Petridis

Received: 8 April 2024
Revised: 23 May 2024
Accepted: 24 May 2024
Published: 28 May 2024

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1. Introduction

Artificial Intelligence (AI) is the development of machines that can imitate human intelligence in thinking, acting, and learning [1]. The main goal of AI is to enable machines to perform tasks that typically require human intelligence, like visual perception, speech recognition, decision making, and language translation. The journey of AI began in the mid-20th century when people began to imagine creating machines with human-like intelligence. Since 1950, when Alan Turing proposed the concept of a universal machine that could perform any mathematical computation [2], AI applications have come a long way, providing a wide range of benefits for personal and professional tasks [3]. AI is used in various personal applications, such as smart assistants like Siri and Alexa, which can manage devices and provide essential information. In entertainment, AI systems suggest personalized content on platforms like Netflix (https://www.netflix.com/, accessed on 2 April 2020).

Generative AI applications like ChatGPT are extensively used by a vast audience and students daily [4], indicating their significant impact on enhancing human–computer interaction [5,6]. In the workplace, AI applications automate and optimize various processes, such as data analysis. Furthermore, AI systems assist in diagnostics, tailor medical devices, and provide healthcare [7].

Over the past two years, higher education students have seen a noticeable increase in the use of AI applications [8]. Many applications, such as ChatGPT, are used for various purposes [9]. This trend has sparked a debate in the international educational community, particularly at the higher-education level [8]. Some academic institutions have even banned AI, a decision that most researchers disagree with [10–12]. Instead, researchers propose that the best solution is to critically integrate AI into the educational process based on a specific framework of norms that includes the integration of AI in teaching practice, learning, and learning assessment [12–17]. Numerous projects propose and develop AI applications to support students' learning in higher education. For instance, augMENTOR (https://augmentor-project.eu/, accessed on 2 April 2020) aims to create a new pedagogical framework that promotes basic skills and 21st-century competencies by integrating emerging technologies. This framework will be supported by an open-access AI-boosted toolkit that builds on big data and learning analytics' strengths. It provides different types of stakeholders with explainable recommendations for the smart search and identification of educational resources, as well as for designing personalized learning profiles that consider individual actors' characteristics, needs, and preferences. augMENTOR will leverage advancements in Pedagogical Design, Creative Pedagogy, Explainable AI, and Knowledge Representation and Reasoning for instructional purposes [18].

In order to effectively use specific applications, it is essential to understand the experiences of students who have used them in the past [8] and to identify the factors that influence their intention to use them in the future [19]. However, limited studies have been conducted on this topic in the European Union, with only a few studies conducted in Greece [20]. Previous studies have primarily used the technology acceptance model (TAM), e.g., [21,22] and, to a lesser degree, the Unified Theory of Acceptance and Use of Technology (UTAUT), e.g., [19,23–26]. However, these studies have investigated the factors influencing the intention to use these applications with participants from various scientific disciplines without specifying any particular field of study. Given that most of these AI applications are used as writing and brainstorming assistants (e.g., to facilitate literature searching and summarizing readings) [4,27,28], it is essential to investigate the factors that influence the intention to use these applications with students who are studying the humanities and social sciences [28].

This study aims to fill the knowledge gap by using the Unified Theory of Acceptance and Use of Technology (UTAUT 2) [29] and investigating the factors influencing the intention and actual use of AI technology among higher education students. Specifically, this study looks into the factors that explain how Greek students in humanities and social sciences use AI applications for academic purposes.

The findings of this research could help improve our understanding of educational systems and provide insights for potential improvements and reforms. These insights could help develop new academic applications that meet the evolving needs of students in a technology-driven landscape. Moreover, faculty members and policymakers could use this research to create a framework for the responsible and effective utilization of these applications to enhance teaching and learning in higher education.

This manuscript is structured as follows: First, we present the students' usage of AI applications, along with the conceptual model UTAUT2 and the corresponding hypotheses. Next, we describe the analytical methodology used in this study. Finally, we present this study's detailed results and discuss them in the following section, including the findings of previous studies.

2. Theoretical Framework

2.1. Students' Use of AI for Academic Purposes

While discussions surrounding AI and its applications in education have persisted for over a decade [30,31], the recent evolution and widespread adoption of AI in the past few years have ignited fresh interest, raised concerns, and sparked debates. AI models now facilitate the generation of multimodal content, encompassing text, images, and videos based on textual prompts. Notable applications include ChatGPT and CoPilot for text-to-text generation, Stable Diffusion, DALL-E, and Midjourney for text-to-image conversion, and DeepBrain AI and Sora for text-to-video generation [32,33]. The potential of AI to revolutionize current practices across various sectors, such as business, education, healthcare, and content generation, is considerable [33]. Large Language Models like ChatGPT, in particular, have seen remarkably rapid adoption by the public, surpassing platforms like Twitter or Facebook [32], prompting extensive research into their implications for education and learning [4].

Studies examining the applications and advantages of ChatGPT, particularly for students in higher education across various disciplines, have highlighted several benefits. Among these are the potential for personalized learning experiences, tailored feedback, and customized learning tasks. Furthermore, ChatGPT has shown promise in reshaping paradigms surrounding student assessment and evaluation, fostering increased motivation and engagement, enhancing communication skills in language acquisition, and aiding students in making informed academic decisions [13,21,34,35].

Moreover, AI applications like ChatGPT are suitable for students across all disciplines, including humanities and social sciences [4,28]. These applications encompass a wide array of functionalities, including language translation, enhanced access to information, the summarization of relevant content tailored to individual queries, question-answering capabilities, support for teaching practices, the facilitation of research tasks such as data analysis and interpretation, and guidance on data collection methods [34,36].

2.2. Technology Acceptance Model of AI

Several models and frameworks have emerged to elucidate user adoption of novel technologies. Venkatesh et al. [37] conducted a comparative analysis of eight models originating from sociology, psychology, and communications. They introduced and empirically validated the UTAUT model, which surpassed the efficacy of the original eight models, providing a robust theoretical foundation by systematically integrating and extending previous models [37]. The Unified Theory of Acceptance and Use of Technology 2 (UTAUT2) model extends the foundational UTAUT model, aiming to provide deeper insights into the factors influencing technology adoption and usage, particularly in consumer contexts. In addition, UTAUT2 offers enhanced explanatory and predictive power compared to earlier models. Venkatesh, Thong, and Xu introduced this extension in their 2012 paper "Consumer Acceptance and Use of Information Technology: Extending the Unified Theory of Acceptance and Use of Technology" [29].

The UTAUT2 model offers a nuanced perspective on how diverse factors such as enjoyment, cost, and habit collectively impact the adoption and sustained use of AI applications within personal contexts. This comprehensive understanding renders it a valuable framework for examining technology usage among students and educators for educational purposes. Including new constructs like hedonic motivation, price value, and habit helps capture a broader range of factors influencing technology acceptance and use. By encompassing both practical and affective dimensions of technology adoption, the UTAUT2 model empowers stakeholders to design proactive interventions tailored to educational settings [37,38]. Finally, numerous empirical studies have validated the UTAUT2 model in AI usage in various populations (see Table 1). This empirical support enhances the model's credibility and reliability as a tool for understanding AI applications' usage by Greek students from humanities and social sciences schools.

Several key constructs are pivotal in understanding adoption and usage behaviours in the context of students' utilization of AI applications. Performance expectancy (PerExp) pertains to the extent to which students anticipate that employing AI applications will yield benefits in carrying out specific tasks. Effort expectancy (EfExp) denotes the perceived ease of utilizing AI applications. Social influence (SocInf) encompasses students' perceptions of the expectations of other students regarding the usage of AI applications. Facilitating conditions (FacCon) gauge the extent to which students believe that the technical and organizational infrastructure is in place to support the utilization of AI applications. Behavioural intention (BehInt) signifies the degree of inclination or intention to utilize AI applications, whereas Use Behaviour (UsBeh) represents the actual engagement with and utilization of AI applications.

Moreover, UTAUT2 introduces three additional constructs tailored to consumer contexts. Habit (Hab) assesses the degree to which students engage in behaviours automatically due to learned routines, emphasizing the significance of past behaviours and experiences in predicting the future utilization of AI applications. Hedonic motivation (HedMot) reflects the enjoyment or pleasure of using AI applications. Price value (PrVal) encapsulates the cognitive assessment of the perceived benefits of the applications weighed against the monetary cost associated with their usage. Furthermore, gender, age, and experience are moderating variables capable of influencing the strength and direction of relationships between the core constructs of the model and outcomes such as behavioural intention (BehInt) and Use Behaviour (UsBeh).

Despite the burgeoning interest in AI in recent years, there remains a need for more research studies on students' utilization of AI worldwide [39]. Recently, however, a handful of studies employing the UTAUT model across various countries have shed light on a consistent set of factors influencing students' behavioural intentions and acceptance of AI in general [19,25], as well as specifically in chatbot technologies [23,24,26]. Some studies have focused solely on explicating the factors underlying students' behavioural intentions towards using AI applications [19,24], with only one study exploring the moderating effect (specifically, the influence of study years and gender) [40]. Moreover, three studies have examined these factors within the context of the European Union, specifically in England [23], Spain [26], and Poland [40].

Table 1 provides an overview of recent studies concerning higher education students' behavioural intention to utilize AI applications for academic purposes. Notably, Alzahrani [19], employing a combination of TAM and UTAUT models with a sample of 350 students from universities in Saudi Arabia, investigated the factors influencing students' behavioural intention to use AI applications in general. Their findings revealed that performance expectancy, effort expectancy, and facilitating conditions significantly impacted behavioural intention. Similarly, Alshammari and Alshammari [24], utilizing the UTAUT model with 136 students from the same country, found that performance expectancy and facilitating conditions were critical determinants of students' behavioural intention to use ChatGPT. Furthermore, Dahri et al. [25], employing the UTAUT model with 305 university students (203 from Pakistan and 102 from Malaysia), identified performance expectancy and facilitating conditions influencing students' behavioural intention to use AI applications. Finally, it is observed that behavioural intention ultimately translates into actual usage behaviour.

In the European Union, Almahri et al. [23] conducted a study with 431 higher education students from the UK, utilizing an adapted UTAUT2 model devoid of moderating effects. Their findings indicated that performance expectancy, expected effort, and habit elucidated students' behavioural intention to use ChatGPT. Furthermore, the behavioural intention was found to correlate with actual usage. Similarly, Romero-Rodríguez et al. [26], examining 400 students from various universities in Spain, employed the UTAUT2 model without moderating effects. Their research revealed that performance expectancy, hedonic motivation, price value, and habit significantly explained students' behavioural intention

to use ChatGPT. Additionally, habit, facilitating conditions, and behavioural intention were identified as predictors of actual usage behaviour.

Furthermore, Strzelecki [40] investigated 534 students from diverse Polish universities using the UTAUT2 model, incorporating moderating effects related to gender and years of study. Their study unveiled that habit, performance expectancy, hedonic motivation, effort expectancy, and social influence contributed to explaining students' behavioural intention to use ChatGPT. Additionally, habit, facilitating conditions, and behavioural intention were associated with actual usage. Notably, no significant moderating effects were observed. Table 1 provides an overview of recent studies focusing on higher education students' behavioural intention to utilize AI applications for academic purposes.

Table 1. Recent studies on higher education students' behavioural intentions towards utilizing AI applications for academic purposes.

Authors	Years of Study	Technology Acceptance Model Used	Sample Size (Nationality of Higher Education Students)	AI App	Significant Effects
Almahri et al. [23]	2020	Adapted UTAUT2 (without moderating effects)	431 (UK)	ChatGPT	PerExp, EfExp, Hab ->> BehInt BehInt ->> UsBeh
Alshammari and Alshammari [24]	2024	UTAUT	136 (Saudi Arabia)	ChatGPT	PerExp, FacCon ->> BehInt
Alzahrani [19]	2023	Adapted UTAUT	350 (Saudi Arabia)	AI in general	PerExp, EfExp, FacCon ->> BehInt
Dahri et al. [25]	2024	UTAUT	305 (203 Pakistan and 102 Malaysia).	AI in general	PerExp, FacCon ->> BehInt (BehInt ->> UsBeh)
Romero-Rodríguez et al. [26]	2023	UTAUT2 (without moderating effects)	400 (Spain)	ChatGPT	PerExp, HedMot, Hab, PrVal ->> BehInt (HedMot, FacCon, BehInt ->> UsBeh)
Strzelecki [40]	2023	UTAUT2 (with moderating effects)	534 (Polish)	ChatGPT	PerExp, EfExp, SocInf, Hab, HedMot, ->>BehInt, (FacCon, Hab, BehInt → UsBeh)

Notes. Use Behaviour (UsBeh), behavioural intention (BehInt), performance expectancy (PerExp), effort expectancy (EfExp), social influence (SocInf), facilitating conditions (FacCon), hedonic motivation (HedMot), price value (PrVal), and habit (Hab).

2.3. Research Hypotheses

Utilizing the UTAUT2 model as the framework, we investigated eleven hypotheses (refer to Figure 1), encompassing both the main effects and interactions influenced by moderating factors such as gender, age, and experience with AI:

Hypothesis 1 (H1): *Expected performance (PerExp) positively influences students' intention to use AI applications for academic purposes (PerExp → BehInt).*

Hypothesis 2 (H2): *Expected effort (EfExp) positively impacts students' intention to use AI applications for academic purposes (EfExp → BehInt).*

Hypothesis 3 (H3): *Social influence (SocInf) positively contributes to students' intention to use AI applications for academic purposes (SocInf → BehInt).*

Hypothesis 4 (H4): *Facilitating conditions (FacCon) positively affect students' intention to use AI applications for academic purposes (FacCon → BehInt).*

Hypothesis 5 (H5): *Hedonic motivation (HedMot) positively influences students' intention to use AI applications for academic purposes (HedMot → BehInt).*

Hypothesis 6 (H6): *Price value (PrVal) positively affects students' intention to use AI applications for academic purposes (PrVal → BehInt).*

Hypothesis 7 (H7): *Habit (Hab) positively impacts students' intention to use AI applications for academic purposes (Hab → BehInt).*

Hypothesis 8 (H8): *Intention to use (BehInt) positively predicts students' use of AI applications for academic purposes (BehInt → UsBeh).*

Hypothesis 9 (H9): *Facilitating conditions (FacCon) positively influence students' use of AI applications for academic purposes (FacCon → UsBeh).*

Hypothesis 10 (H10): *Habit (Hab) positively influences students' use of AI applications for academic purposes (Hab → UsBeh).*

Hypothesis 11 (H11): *Gender, age, and experience are moderating factors influencing the utilization of AI applications for academic purposes.*

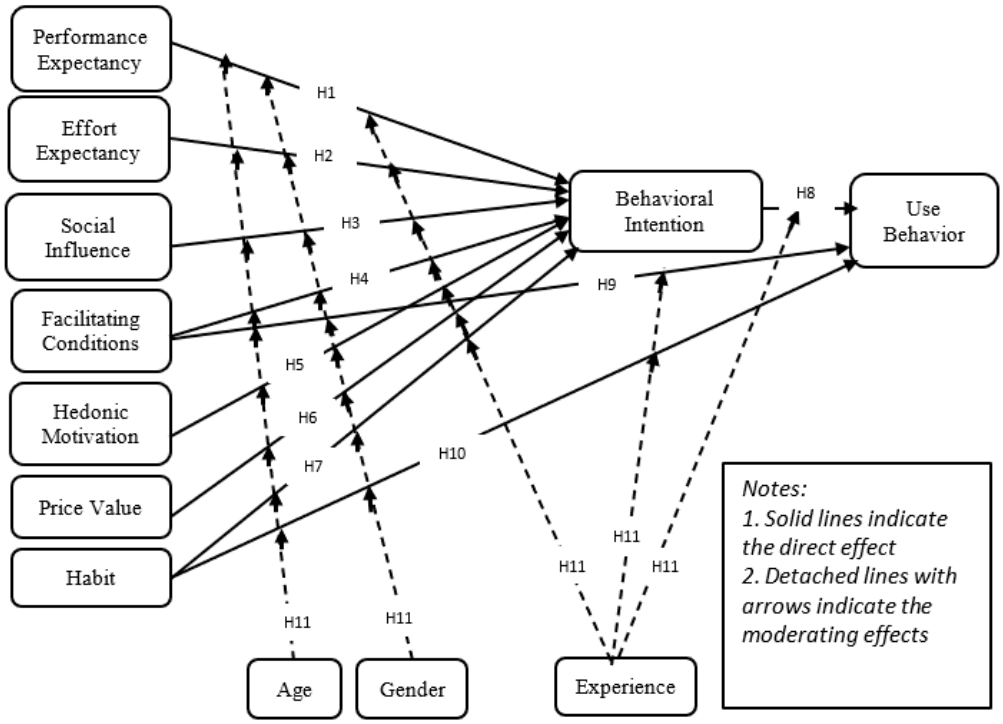

Figure 1. Direct effects among the constructs of the UTAUT2.

3. Research Methods

This study was conducted during November and December 2023, following approval from the Institutional Review Board of the Department of Educational Science and Early Childhood Education at the University of Patras (85812/09-11-2023). This cross-sectional investigation employed a quantitative educational research strategy, with data collection facilitated through an online questionnaire [41]. Participants were given one month to complete the questionnaire, with the survey closing at the end of December 2023.

3.1. Research Instrument

The survey questionnaire comprised two sections, incorporating an objective overview, completion instructions, and a guarantee of respondent anonymity [42]. The first section focused on gathering demographic information from student participants, including gender, age, year of study, and frequency of experience with AI applications for academic purposes. The second section comprised 27 statements aligning with the nine constructs of the UTAUT2 model. The responses were recorded on a five-point scale ranging from 1 (strongly disagree) to 5 (strongly agree). These statements, three for each construct, were adapted

from a validated and reliable research instrument utilized by Nikolopoulou et al. [38]. Their study employed the UTAUT2 model to explore factors influencing Greek higher education students' use of mobile devices for academic purposes [38]. Specifically, the adaptation involved substituting "mobile devices" with "AI applications" (refer to the Appendix A for corresponding statements for each construct). This final version of the instrument was based on the results of the pilot administration of the instrument with five students (who were excluded from the final sample). The wording in some items was revised considering the difficulties and ambiguities in interpreting that were declared during an interview with this pilot sample's students. Moreover, in the measurement model section, we provide proof of the research instrument's psychometric properties (validity and reliability).

3.2. The Strategy of Data Analysis

For data analysis, the R environment [43] along with the "seminr" package [44] was employed. Structural equation modeling (Partial Least Squares-Structural Equation Modeling or "PLS-SEM") was utilized as the method, enabling the estimation of intricate cause–effect relationships in path models featuring latent variables [45]. This method was deemed appropriate for the UTAUT2 model due to its complexity, incorporating numerous constructs (nine) and moderator variables (three), along with indicators (27 statements) and model relationships. Additionally, PLS-SEM is well suited for studies with small sample sizes yet complex models, such as the present research [46]. To present the results, guidelines outlined by Hair et al. [46] were followed.

The bootstrapping method was employed to estimate parameters, including path coefficients and their confidence intervals, for both the measurement and structural models. This involved generating 2000 random samples with replacements from the original dataset [45]. Lastly, the measurement model addressed psychometric properties such as convergent and discriminant validity, as well as the reliability of the research instrument in detail.

3.3. Participants

The convenience sample for this study comprised 197 students from diverse departments within the School of Humanities and Social Sciences at the University of Patras. These departments included Philosophy (62 students), Educational Sciences and Early Childhood Education (102 students), and Philology (33 students). With the assistance of academic teachers, a subset of students from each department was selected, focusing on those taught by the respective faculty members. Invitations to participate in the survey were sent via email by the faculty members to the selected students. Table 2 provides an overview of the demographic characteristics of these participants. According to the distributions of gender in this department (in the department of Educational Sciences and Early Childhood Education, 95% are female; in the departments of Philosophy and Philology, 20% are female), the over-representation of female participants does not indicate any sampling bias.

Table 2. Demographic profile of participating students in the School of Humanities and Social Sciences (N = 197).

	Frequencies	**Relative Frequencies (%)**
Gender		
Male	25	12.7
Female	172	87.3
Age		
up to 19	131	66.5
20–21	52	26.4
at least 22	14	7.1

Table 2. *Cont.*

	Frequencies	Relative Frequencies (%)
Years of study		
First	85	43.2
Second	40	20.4
Third	50	25.3
Fourth	15	7.6
Last	7	3.5
Experience with AI		
Never	79	40.1
Sometimes, in a month	85	43.1
Many times a week	23	11.7
Daily	10	5.1

4. Results

First, the measurement model will be presented, offering insights into the reliability and validity of the research instrument. Subsequently, the structural model, which entails testing the hypotheses of the conceptual model, will be examined.

4.1. Measurement Model

Table 3 presents descriptive statistics, reliability coefficients, and indices of convergent validity for each construct of the UTAUT2 model. The Cronbach's Alphas for all constructs are either close to or exceed 0.7, while composite reliability scores surpass 0.7, indicating satisfactory internal consistency reliability [45]. Additionally, item loadings are predominantly above 0.7, and the Average Extracted Variance for each construct exceeds 0.5, suggesting a satisfactory level of convergent validity [45].

Table 3. Descriptive statistics, reliability, and validity indices for the constructs of the UTAUT2 model.

	Mean (SD.)	λ	α	CR	AVE
Use Behaviour (UsBeh)	2.86 (0.77)		0.706	0.838	0.636
UsBeh1		0.803			
UsBeh2		0.906			
UsBeh3		0.666			
Behavioural Intention (BehInt)	3.00 (0.95)		0.923	0.951	0.866
BehInt1		0.923			
BehInt2		0.920			
BehInt3		0.949			
Performance Expectancy (PerExp)	3.37 (0.82)		0.865	0.918	0.848
PerExp1		0.903			
PerExp2		0.893			
PerExp3		0.866			
Effort Expectancy (EfExp)	3.81 (0.88)		0.914	0.943	0.829
EfExp1		0.894			
EfExp2		0.942			
EfExp3		0.926			
Social Influence (SocInf)	2.58 (0.87)		0.902	0.938	0.837
SocInf1		0.902			
SocInf2		0.938			
SocInf3		0.904			
Facilitating Conditions (FacCon)	3.32 (0.75)		0.669	0.807	0.584
FacCon1		0.690			
FacCon2		0.836			
FacCon3		0.759			
Hedonic Motivation (HedMot)	3.24 (0.79)		0.809	0.886	0.723
HedMot1		0.790			
HedMot2		0.862			
HedMot3		0.895			

Table 3. *Cont.*

	Mean (SD.)	λ	α	CR	AVE
Price Value (PrVal)	3.06 (0.71)		0.833	0.898	0.748
PrVal1		0.722			
PrVal2		0.922			
PrVal3		0.934			
Habit (Hab)	2.65 (0.79)		0.657	0.815	0.595
Hab1		0.827			
Hab3		0.723			
Hab4		0.760			

Notes: SD = Standard Deviation, λ = Factor Loadings, α = Cronbach's Alpha, CR = composite reliability, and AVE = average variance extracted.

Furthermore, Pearson's linear correlation coefficients among the constructs are statistically significant (see Table 4). Additionally, the Fornell–Larcker criterion [47] demonstrates the satisfactory discriminant validity of the constructs, as evidenced by the higher values of the square roots of the average variance extracted (presented in the diagonal cells) compared to all inter-construct correlations.

Table 4. Product Moment Pearson's linear correlation coefficients and Fornell-Larcker discriminant validity criterion for the UTAUT2 constructs.

	1	2	3	4	5	6	7	8	9
1. UsBeh	(0.797)								
2. BehInt	0.738 **	(0.930)							
3. PerExp	0.614 **	0.703 **	(0.921)						
4. EfExp	0.179 *	0.207 **	0.450 **	(0.910)					
5. SocInf	0.498 **	0.559 **	0.489 **	0.044	(0.915)				
6. FacCon	0.434 **	0.388 **	0.502 **	0.462 **	0.352 **	(0.764)			
7. HedMot	0.601 **	0.591 **	0.636 **	0.384 **	0.360 **	0.514 **	(0.850)		
8. PrVal	0.393 **	0.418 **	0.451 **	0.314 **	0.318 **	0.379 **	0.554 **	(0.865)	
9. Hab	0.720 **	0.714 **	0.548 **	0.147 *	0.559 **	0.334 **	0.516 **	0.459 **	(0.771)

Notes: correlation is significant at the **. 0.01 level *. 0.05 (two-tailed). The diagonal cells represent the square roots of the average variance extracted (AVE) for each construct, while the lower triangles display the correlations among the constructs. Use Behaviour (UsBeh), behavioural intention (BehInt), performance expectancy (PerExp), effort expectancy (EfExp), social influence (SocInf), facilitating conditions (FacCon), hedonic motivation (HedMot), price value (PrVal), and habit (Hab).

4.2. Structural Model

The analysis revealed no collinearity issues, as all Variance Inflation Factor (VIF) coefficients were below 3 [46]. The explained variance in the two endogenous constructs (R^2 of BehInt = 75% and R^2 of UsBeh = 67%) indicates moderate to substantial predictive power [45].

Only six of the eleven hypotheses were supported in the testing (refer to Table 5), as indicated by excluding zero from the corresponding 95% confidence intervals. In the confirmed hypotheses, the direct effect coefficients suggest small to medium effects [45]. Notably, the hypothesis regarding the moderating effects of gender, age, and experience with AI applications was not confirmed.

Table 5. Testing the Assumptions of the conceptual model: direct effect coefficients (b) and their 95% confidence intervals based on bootstrapping (2000 samples).

Hypotheses		Direct Effect	95% CI		Results
H1	PerExp → BehInt	0.422	0.232	0.571	Supported
H2	EfExp → BehInt	−0.058	−0.176	0.084	Not supported
H3	SocInf → BehInt	0.081	−0.025	0.217	Not supported
H4	FacCon → BehInt	0.010	−0.114	0.138	Not supported

Table 5. *Cont.*

Hypotheses		Direct Effect	95% CI		Results
H5	HedMot → BehInt	0.184	0.052	0.315	Supported
H6	PrVal → BehInt	−0.034	−0.153	0.108	Not supported
H7	Hab → BehInt	0.335	0.187	0.488	Supported
H8	BehInt → UsBeh	0.423	0.262	0.558	Supported
H9	FacCon → UsBeh	0.187	0.078	0.292	Supported
H10	Hab → UsBeh	0.284	0.151	0.420	Supported
H11	Moderating effects		Not supported		

Notes: Use Behaviour (UsBeh), behavioural intention (BehInt), performance expectancy (PerExp), effort expectancy (EfExp), social influence (SocInf), facilitating conditions (FacCon), hedonic motivation (HedMot), price value (PrVal), and habit (Hab). BehInt (R^2 = 75%); UsBeh (R^2 = 67%).

5. Discussion of Results

In this study, utilizing a research instrument comprising 27 statements, we delved into the factors outlined in the UTAUT2 technology acceptance model to elucidate the intention and actual utilization of AI applications among humanities and social sciences students for academic purposes. Our findings affirm this research instrument's satisfactory validity and reliability within the current sample of Greek students from the School of Humanities and Social Sciences. Additionally, the data analysis underscores the robust structure of the UTAUT2 model and its adequate fit with the data about factors explaining the intention of higher education students to utilize AI applications [23,26,40]. Moreover, as reflected in the percentage of explained variance concerning students' intention and use of AI applications, the model's explanatory power aligns with previous research findings [26,40].

Specifically, in elucidating students' intention to use AI applications, performance expectancy (direct effect = 0.422), habit (direct effect = 0.335), and hedonic motivation (direct effect = 0.184) emerge as dominant factors. These positive effects indicate that heightened perceptions among students regarding the performance expectancy of AI applications in supporting their academic endeavours, when other factors in the model are held constant, are more likely to lead to their future utilization. Similarly, a more robust perception of specific applications becoming habitual and the pleasure derived from their utilization positively influence their likelihood of use. Notably, previous research has highlighted the explanatory role of students' attitudes towards the expected performance of AI applications [19,23–26,40], as well as the significance of habit [23,26,40] and hedonic motivation [26,40].

In explaining the actual use of AI applications by students, factors such as intention to use (direct effect = 0.423), habit (direct and indirect effect = 0.425), and facilitating conditions (direct and indirect effect = 0.191) emerge as pivotal. These positive effects indicate that strong student perceptions regarding their intention to use AI applications for academic support are more likely to translate into actual utilization when other factors in the model are held constant. Additionally, the perception of specific applications becoming habitual and the presence of facilitating conditions further bolster their actual usage.

Prior studies have also observed similar effects [23,25,26,40]. Furthermore, the stronger the perception among students that these applications have become habitual, the more inclined they are to integrate them into their academic routines. This finding has also been consistently reported in prior research [26,40]. Similarly, albeit to a lesser degree, students are more likely to adopt these applications for academic support when they perceive adequate technical support. This modest effect has also been documented in previous studies [26,40], often attributed to the ease of use of these applications, which are typically similar to other digital tools commonly used by students.

Notably, effort expectancy did not emerge as a significant predictor of students' intention in this study. Additionally, other research corroborates the absence of moderation effects, as posited in the UTAUT2 model [40]. However, it is essential to highlight that this particular study [40] only considered gender and years of study as moderating factors, without taking into account experience with AI applications. However, it is essential to rec-

ognize that previous findings were based on participants from higher education institutions in general rather than specifically from the schools of humanities and social sciences. Moreover, it is worth mentioning that this sample might be relatively homogeneous regarding gender and experience. If the participants predominantly share similar levels of experience (in this sample, most of the sample has "Never or Sometimes in a month" experience with AI applications), or if gender distribution is imbalanced (in this sample, the considerable majority of the sample is female), it can be challenging to detect moderating effects [45].

Implications and Limitations

Given the insights above, several practical applications and avenues for future research emerge. The determinants identified regarding higher education students' intentions to adopt AI applications suggest that faculty members and policymakers can leverage this model to cultivate learning environments conducive to AI utilization, benefiting students across various disciplines, including those from humanities and social sciences schools. However, practitioners must also address ethical considerations such as plagiarism and the potential for excessive reliance on this technology, which may lead to inappropriate usage [10–12,17].

Academics and lecturers should elucidate and demonstrate the utility of AI applications as supportive tools for students in humanities and social sciences schools, irrespective of their year of study or prior experience with such applications. This entails integrating demonstrations of these applications' benefits and advantages into courses, student seminars, and technology-focused workshops, thereby enhancing students' efficiency [34] and academic performance [25]. For example, the students could use AI applications that offer personalized learning and advice based on their academic needs. Additionally, students could use AI applications incorporating game elements and interactive learning environments with quizzes to make learning more fun and engaging. Finally, students should integrate AI applications into their daily learning routines. Within this framework, presentations could encompass implementation strategies, fundamental functions, and engaging features of AI applications. Moreover, efforts could be made to incorporate these applications into existing curricula.

However, in light of the potential for students to develop an over-reliance on these applications due to habit formation and enthusiastic adoption, academic staff should advocate for the critical integration of AI into the educational process. This should be guided by a framework of established norms and entail the thoughtful integration of AI into teaching practices, learning activities, and assessment methodologies [17].

Lastly, creators of AI applications should prioritize providing technical support in the form of feedback mechanisms and clear instructions, facilitating users' ability to address any challenges encountered during utilization effectively.

As with any research endeavour, it is essential to acknowledge certain limitations when interpreting the findings, which also pave the way for future research avenues. Firstly, there exists an apparent potential bias in the results due to the over-representation of female students. However, it is noteworthy that female students comprise the majority within Greece's School of Humanities and Social Sciences. In addition, considering the convenience sampling, this sample needs to be more representative. Moreover, the effects observed between variables should primarily be considered correlations, since this study adopts a cross-sectional approach [41].

Furthermore, the reliance on self-report measures in the study increases the likelihood of measurement bias and socially desirable responses [48]. Future longitudinal studies are warranted to address these limitations. Such studies could track changes over time, allowing researchers to observe shifts in students' intentions and usage of AI applications. These longitudinal investigations should strive for a representative sample, potentially employing a cluster sampling technique [41] encompassing students from diverse departments and universities to enhance the generalizability of the results.

Additionally, to comprehensively understand the factors influencing both intention and actual use of AI applications, it is imperative to explore other variables such as perceived trust [21], information accuracy [25], and perceived satisfaction [22,25]. Combining UTAUT2 with these factors could provide a more comprehensive understanding of students from humanities and social sciences schools and their intention to use AI applications for academic purposes. Finally, supplementing quantitative analysis with a qualitative approach involving semi-structured interviews could yield rich insights in relation to the factors above.

Author Contributions: Conceptualization, K.L., V.K., I.V. and N.K.; Methodology, K.L., S.P., S.A. and V.K.; Software, A.F. and S.A.; Validation, K.L., S.P. and N.K.; Formal analysis, K.L. and S.A.; Investigation, K.L.; Resources, S.P., A.F. and A.A.; Data curation, K.L., S.A. and A.A.; Writing—original draft, K.L. and V.K.; Visualization, N.K.; Supervision, K.L., V.K. and N.K.; Project administration, K.L. and S.P. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: This study was conducted in accordance with the Declaration of Helsinki and approved by the Institutional Review Board (or Ethics Committee) of the Department of Educational Science and Early Childhood Education of the University of Patras (85812/09-11-2023).

Informed Consent Statement: Informed consent was obtained from all subjects involved in this study.

Data Availability Statement: The datasets generated during and analysed during the current study are available from the first author upon reasonable request.

Acknowledgments: The authors are indebted to the students of the university departments for their continued support. This research was carried out in the context of the European Union Horizon 2020 project augMENTOR ("Augmented Intelligence for Pedagogically Sustained Training and Education", HORIZON-CL2-2021-TRANSFORMATIONS-01-05), co-funded by the European Commission under Grant Agreement ID: 101061509.

Conflicts of Interest: The authors declare no conflicts of interest.

Appendix A

1. PerExp1. I find Artificial Intelligence applications such as ChatGPT useful in my studies.
2. PerExp2. Using Artificial Intelligence applications like ChatGPT helps me to complete various activities related to my studies faster
3. PerExp3. Using Artificial Intelligence applications like ChatGPT increases productivity in my studies
4. EfExp1. It is easy to learn using Artificial Intelligence applications like ChatGPT.
5. EfExp2. My interaction with Artificial Intelligence applications like ChatGPT is clear and understandable
6. EfExp3. Artificial Intelligence applications like ChatGPT are easy to use.
7. SocInf1. The people who are important to me (e.g., friends and family) believe I should use Artificial Intelligence applications like ChatGPT for my studies.
8. SocInf2. The people influencing my behaviour believe I should use Artificial Intelligence applications like ChatGPT in my studies.
9. SocInf3. The people whose opinions I value prefer that I use Artificial Intelligence applications, like ChatGPT, for my studies as well
10. FacCon1. It is easy to find user instructions for Artificial Intelligence applications like ChatGPT.
11. FacCon2. I have the knowledge necessary to use Artificial Intelligence applications like ChatGPT.
12. FacCon3. The Artificial Intelligence applications like ChatGPT that I use in my studies align well with other applications I use
13. HedMot1. The use of Artificial Intelligence applications like ChatGPT in my studies is enjoyable.

14. HedMot2. The use of Artificial Intelligence applications like ChatGPT in my studies is pleasant.
15. HedMot3. Using Artificial Intelligence applications like ChatGPT in my studies is very entertaining.
16. PrVal1. The cost of Artificial Intelligence applications like ChatGPT is reasonable
17. PrVal2. The cost of the services I access through Artificial Intelligence applications like ChatGPT is worth the money.
18. PrVal3. Artificial Intelligence applications like ChatGPT are worth their cost.
19. Hab1. Using Artificial Intelligence applications like ChatGPT has become a habit for me.
20. Hab2. I must use Artificial Intelligence applications like ChatGPT.
21. Hab3. Using Artificial Intelligence applications like ChatGPT is self-evident for me.
22. UsBeh1. I intend to continue using Artificial Intelligence applications like ChatGPT in my studies.
23. UsBeh2. I will always strive to use Artificial Intelligence applications like ChatGPT in my studies.
24. UsBeh3. I plan to use Artificial Intelligence applications like ChatGPT frequently in my studies.
25. BehInt1. Using Artificial Intelligence applications like ChatGPT is a pleasant experience.
26. BehInt2. I use Artificial Intelligence applications like ChatGPT to support my studies.
27. BehInt3. I spend much time using Artificial Intelligence applications like ChatGPT in my studies

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